SIMULATOR
EXAMINATION SCENARIO GUIDE

SCENARIO TITLE:	15-01 NRC ESG-2
SCENARIO NUMBER:	15-01 NRC ESG-2
EFFECTIVE DATE:	See Approval Dates
EXPECTED DURATION:	60 minutes
REVISION NUMBER:	00
PROGRAM:	[ ] **L.O. REQUAL**
	[X] **INITIAL LICENSE**
	[ ] **STA**
	[ ] **OTHER** ____________

Revision Summary
New issue for 15-01 ILOT NRC exam

PREPARED BY:	G Gauding **Lead Regulatory Exam Author**	8-3-16 **Date**
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SCAN OF SIGNED SCENARIO COVER SHEET

I. OBJECTIVES

A. Given a loss of a heater drain pump or condensate pump, take corrective action IAW S2.OP-AB.CN-0001.

B. Given the order or indications of a feedwater or condensate system malfunction, perform actions as the nuclear control operator to RESPOND to the malfunction in accordance with S2.OP-AB.CN-0001.

C. Given indication of a feedwater or condensate system malfunction, DIRECT the response to the malfunction in accordance with S2.OP-AB.CN-0001.

D. Given a steam generator tube leak, take corrective action, IAW S2.OP-AB.SG-0001.

E. Given the order or indications of a steam generator tube leak (SGTL), perform actions as the nuclear control operator to RESPOND to the tube leak in accordance with S1/S2.OP-AB.SG-0001.

F. Given the order or indications of a steam generator tube leak (SGTL), DIRECT the response to the tube leak, in accordance with S2.OP-AB.SG-0001.

G. Given the order or indications of a reactor trip, perform actions as the nuclear control operator to RESPOND to the reactor trip in accordance with 2-EOP-TRIP-1.

H. Given indication of a reactor trip, DIRECT the response to the reactor trip in accordance with 2-EOP-TRIP-1.

I. Given the order or indications of a safety injection, perform actions as the nuclear control operator to RESPOND to the safety injection in accordance with the approved station procedures.

J. Given indication of a safety injection, DIRECT the response to the safety injection in accordance with the approved station procedures.

K. Given the order or indications of a steam generator tube rupture (SGTR), perform actions as the nuclear control operator to RESPOND to the tube rupture in accordance with the approved station procedures.

L. Given indication of a steam generator tube rupture (SGTR), DIRECT the response to the SGTR in accordance with the approved station procedures.

II. MAJOR EVENTS

A. 2C EDG Pre-lube pump failure (TS)

B. 23 Condensate pump trip /Power Reduction

C. 21 SGTL / 21 SGTR

D. Loss of Spray during RCS depressurization

E. Stuck open PZR PORV during RCS depressurization

III. SCENARIO SUMMARY

A. The crew will take the watch with the unit at 89.4% power, BOL. Power was reduced last shift in preparation for performing Main Turbine Valve testing later this shift. 2C EDG Jacket Water heater failed last shift. Condensate Polisher is in service.

B. After the crew assumes the watch, the 2C EDG Pre-lube pump will fail. The CRS will identify 2C EDG is now inoperable, and enter the appropriate LCO. The CRS will initiate actions to ensure S2.OP-ST.500-0001, Electrical Power Systems AC Sources Alignment is performed within one hour.

C. After the EDG failure has been addressed, 23 Condensate pump will trip. The crew will enter S2.OP-AB.CN-0001, Main Feedwater / Condensate System Abnormality, and perform actions to bypass the polisher and feedwater heaters strings as required to maintain SGFP suction pressure.

D. The crew will perform a power reduction to $\leq$ 85% power and ensure SGFP suction pressure remains above 320 psig.

E. During the power reduction, a SGTL will occur on 21 SG. The crew will enter S2.OP-AB.SG-0001, Steam Generator Tube Leak. The crew will perform actions to minimize the spread of secondary contamination and minimize the potential for steam release from the affected SG. The crew will isolate the steam supply to 23 AFW pump from 21 SG, and the CRS will identify 23 AFW is inoperable and enter the appropriate LCO.

F. After the AFW pump LCO is identified, the SG tube will rupture. The crew will identify the SGTR, and initiate a Rx trip and Safety Injection based on the uncontrolled rise in 21 SG NR or WR level.

G. The crew will perform plant stabilization and diagnostics in EOP-TRIP-1, Reactor Trip or Safety Injection. 21 AFW pump will trip shortly after starting. The crew will isolate AFW flow to and steam flow from 21 SG. (CT-1).

H. The crew will transition to EOP-SGTR-1, Steam Generator Tube Rupture to address the tube rupture. The crew will perform a cooldown to target temperature and demonstrate the ability to maintain RCS temperature < Target temperature (CT-2). The crew will initiate a RCS depressurization. During the depressurization the RCP providing driving head for Spray flow will trip. The crew will open a PZR PORV to continue the depressurization. If 2PR1 is the selected PORV, it will not open, and 2PR2 will be used.

I. The PZR PORV opened to depressurize the RCS will fail to shut when demanded after the depressurization is complete. The crew will attempt to terminate the depressurization by closing the PORV Block valve, which will not shut.

J. The scenario will be terminated after the CRS transitions to EOP-SGTR-3, SGTR with LOCA – Subcooled Recovery, to address the loss of RCS pressure with the stuck open PORV.

IV. INITIAL CONDITIONS

____ IC-232

PREP FOR TRAINING (i.e. computer setpoints, procedures, bezel covers ,tagged equipment)

Initial	Description
____ 1	VC1and VC4 C/T
____ 2	RCPs (SELF CHECK)
____ 3	RTBs (SELF CHECK)
____ 4	MS167s (SELF CHECK)
____ 5	500 KV SWYD (SELF CHECK)
____ 6	SGFP Trip (SELF CHECK)
____ 7	23 CV PP (SELF CHECK)
____ 8	Complete Attachment 2 “Simulator Ready-for-Training/Examination Checklist.”

Note: Tables with blue headings may be populated by external program, do not change column name without consulting Simulator Support group

EVENT TRIGGERS:

Initial	ET #	Description	
	1	EVENT ACTION:	MONP254 <10. //CONTROL BANK C GROUP POSITION
		COMMAND:	
		PURPOSE:	<update as needed>
	3	EVENT ACTION:	KB202LCI //2PR7 RELIEF VALVE-22 STOP VALVE
		COMMAND:	
		PURPOSE:	<update as needed>

MALFUNCTIONS:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Severity
______ 01	AN0266 SER 266 FAILS - :J20 2C DIESEL GENERATOR URGENT TROUBLE	N/A	N/A	N/A	RT-1	SER POINT FAILS/OVRD TO ON
______ 02	CN0117C 23 CONDENSATE PUMP TRIP	N/A	N/A	N/A	RT-3	
______ 03	SG0078A 21 STEAM GENERATOR TUBE RUPTURE	N/A	N/A	N/A	RT-5	5
______ 04	AN3735 AAS 735 FAILS - :21 TGA SUMP LEVEL HIGH	00:03:00	N/A	N/A	RT-11	AAS POINT FAILS/OVRD TO ON
______ 05	AN3736 AAS 736 FAILS - :22 TGA SUMP LEVEL HIGH	00:03:10	N/A	N/A	RT-11	AAS POINT FAILS/OVRD TO ON
______ 06	AN3737 AAS 737 FAILS - :23 TGA SUMP LEVEL HIGH	00:03:20	N/A	N/A	RT-11	AAS POINT FAILS/OVRD TO ON
______ 07	AN3738 AAS 738 FAILS - :24 TGA SUMP LEVEL HIGH	00:03:30	N/A	N/A	RT-11	AAS POINT FAILS/OVRD TO ON
______ 08	AN3739 AAS 739 FAILS - :25 TGA SUMP LEVEL HIGH	00:03:50	N/A	N/A	RT-11	AAS POINT FAILS/OVRD TO ON
______ 09	AF0181A 21 AUX FEEDWATER PUMP TRIP	00:04:00	N/A	N/A	ET-1	
______ 10	VL0297 2PR1 Fails to Position (0-100%)	N/A	N/A	N/A	N/A	0
______ 11	VL0298 2PR2 Fails to Position (0-100%)	N/A	N/A	N/A	RT-13	100
______ 12	RC0003C 23 RC PUMP ELECTRICAL TRIP	N/A	N/A	N/A	RT-7	
______ 13	RC0003A 21 RC PUMP ELECTRICAL TRIP	N/A	N/A	N/A	RT-9	
______ 14	VL0123 2PR7 Fails to Position (0-100%)	N/A	N/A	N/A	ET-3	90
______ 15	CV0035 CHRG MASTER FLO CNTRLR FAILS H/L	N/A	N/A	N/A	RT-15	26

REMOTES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition
______ 01	SP01A1 Prevailing Wind Speed	N/A	N/A	N/A	N/A	10

_____ 02	SP01A2 Prevailing Wind Direction	N/A	N/A	N/A	N/A	160

OVERRIDES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition/ Severity
_____ 01	C201 A LO QC201AY3 2C DIESEL GEN-TROUBLE ALARM	N/A	N/A	N/A	N/A	ON

OTHER CONDITIONS:

	Description
_____	1. None

V. SEQUENCE OF EVENTS

A. State shift job assignments.

B. Hold a shift briefing, detailing instruction to the shift: (provide crew members a copy of the shift turnover sheet).

C. Inform the crew "The simulator is running. You may commence panel walkdowns at this time. CRS please inform me when your crew is ready to assume the shift".

D. Allow sufficient time for panel walk-downs. When informed by the CRS that the crew is ready to assume the shift, ensure the simulator is cleared of unauthorized personnel.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
1. 2C EDG Pre-Lube pump failure			
Simulator Operator: Insert RT-1 on direction from Lead Evaluator. **MALF: AN0266 SER 266 Fails – J20 2C DIESEL GENERATOR URGENT TROUBLE**			
	PO announces unexpected OHA J-20, 2C DG URGENT TRBL.		
	The crew refers to the ARP and dispatches an operator to investigate.		
Role play: 2 minutes after being dispatched call control room and report 2C EDG Alarm Window A-7 PRE-LUBE PUMP FAILURE is in alarm. You have checked the Pre-lube pump and it is NOT running. The Pre-lube pump breaker remains shut.			
	CRS refers to S2.OP-SO.DG-0003, 2C Diesel Generator Operation and recognizes 2C EDG is now inoperable with the Jacket Water Heater and Pre-lube pump inoperable, and enters LCO 3.8.1.1.b, action b.		
	CRS determines that S2.OP-ST.500-0001, Electrical Power Systems AC Sources Alignment must be performed within one hour		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	to demonstrate the operability of the independent AC Sources (action b.1)		
Proceed to next event on direction from Lead Evaluator.			
2. 23 Condensate pump trip / downpower			
Simulator Operator: Insert <u>**RT-3**</u> on direction from Lead Evaluator. **MALF:CN0117C 23 Condensate pump trip**			
	PO announces 23 Condensate pump trip.		
	CRS enters S2.OP-AB.CN-0001, Main Feedwater / Condensate System Abnormality.		
	PO reports no SGFP tripped.		
	PO reports 23 Condensate pump tripped.		
	PO bypasses condensate polisher by opening 21 thru 23 CN108 if SGFP suction pressure is <320 psig.		
Role Play: If dispatched, after 3 minutes report as NEO that you can't see anything wrong with 21 condensate pump. If dispatched to 4KV breaker, report after 2 minutes that there is an overcurrent relay flag dropped.			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	PO reports SGFP suction pressure.		
Note: Crew should discuss power increase expected when the CN47 is opened. (Note in AB.CN-1).			
	CRS directs 2CN47 opened if suction pressure remains < 320 psig.		
	PO isolates letdown by closing 21-24BG4 and 21-24GB185 if 2CN47 is open.		
	CRS directs power reduction to 85% at ≤5%/min with 2 Condensate pumps and 3 Heater Drain pumps in service IAW Attachment 4.		
	CRS enters S2.OP-AB.LOAD-0001, Rapid Load Reduction, to perform the load reduction.		
	CRS directs initiation of AB.LOAD CAS.		
	RO calculates boron addition and rate for power reduction.		
	RO initiates boration.		
	PO initiates turbine load reduction to 85% at rate directed by CRS.		
3. Master Flow Controller failure			
Simulator Operator: Once the			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
power reduction is underway, insert **RT-15** to fail the Master Flow Controller. **MALF: CV0035 CHRG Master Flow Controller fails H/L** **Severity: 26**			
	RO reports unexpected low seal injection flow alarms on all RCP's.		
	RO reports Master Flow Controller is failed based on lower than expected Charging flow (75 gpm vs 87 gpm)		
	CRS enters S2.OP-AB.CVC-0001, Loss of Charging.		
	RO may use Alarm Response Procedure initially in response to Low Seal Injection Flow to adjust 2CV71 to restore seal injection flow to above the alarm setpoint, but this will not restore normal charging flow.		
	RO reports: - A charging pump is running with no indication of cavitation - RO reports neither a PZR level nor a VCT level channel has failed. - No indication of charging system leak		
	RO reports the Charging Master Flow Controller has failed.		
	RO takes manual control of 23 charging		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	pump, and adjusts speed to maintain PZR level on program IAW Att. 2.		
	RO reports PZR level can be maintained stable or rising.		
	RO announces auto rod motion when it occurs.		
Proceed to next event on direction from Lead Evaluator.			
4. 21 SG Tube Leak			
Simulator Operator: Insert <u>**RT-5**</u> on direction from Lead evaluator. **MALF: SG00078A 21 STEAM GENERATOR TUBE RUPTURE** **Severity: 5**			
	RO reports unexpected OHA A-6 and identifies 2R53A 21 Main Steamline N2 monitor from CRT indication in alarm.		
	RO reports 2R53A level >1000 gpd.		
	CRS enters S2.OP-AB.SG-0001, Steam Generator Tube Leak, and S2.OP-AB.RAD-0001, Abnormal Radiation.		
	RO reports OHA A-6 reflash is 2R15 Condenser Air Ejector monitor in alarm.		
	CRS directs initiation of AB.SG CAS.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO reports when 2R19A is in warning and alarm.		
	PO reports when SGBD isolation occurs on high radiation.		
	CRS dispatches operator to isolate TGA and Condensate Polisher area sump pumps.		
Simulator Operator: When dispatched to isolate TGA and polisher building sumps, insert **RT-11**. This deenergizes 21-25 TGA sump pumps after a 3 minute time delay.			
	RO reports PZR level being maintained stable by rising charging flow.		
Note: Initial charging flow with stable PZR level is ~89 gpm. 23 charging pump will deliver ~99 gpm at full flow and seal injection of ~26 gpm. **IF** RO reports PZR level is lowering (due to lag in Master Flow control response,) then RO will swap to a centrifugal charging pump at Step 3.6.			
	RO reports PZR level can be maintained stable or rising.		
Note: SGBD will isolate ~2.5 minutes after 2R53D alarms. This			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
will cause RCS temp and PZR level to rise.			
	PO reports when load reduction is complete.		
	RO reports unit 2 is in MODE 1.		
	RO reports 21 SG is affected SG.		
	PO sets 21MS10 to 1045 psig, and ensures 21GB4, 21MS18, and 21MS7 are shut.		
	CRS dispatches an operator to shut 21MS45 steam supply to 23 AFW pump.		
Note: Do **<u>NOT</u>** report the 21MS45 is shut until AFTER 23 AFW pump has been tripped at step 4.7 of SGTR-1.			
	CRS enters LCO 3.7.1.2 for less than 3 operable AFW pumps, and LCO 3.4.7.2.c for >150 gpd primary to secondary leakage though 21 SG.		
	CRS determines (Action Level 3) the unit must be less than 50% in less than one hour, the unit placed in Hot Standby within 6 hours.		
	CRS dispatches an operator to realign SGBD and Sampling.		
Proceed to next event on direction from Lead Evaluator after Tech Specs have been addressed.			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
5. SGTR			
Simulator Operator: MODIFY MALF SG00078A from 5 to 650 on direction from Lead Evaluator.			
	RO reports indications of worsening tube leak/rupture, with lowering PZR level and pressure.		
	CRS directs RO to trip the Rx, confirm the Rx trip, and initiate a Safety Injection, then enter TRIP-1.		
	RO trips the Rx, confirms the Rx trip, and initiates Safety Injection.		
Simulator Operator: Ensure **ET-1** is true when the rx is tripped. This trips 21 AFW pump after a 4 minute delay.			
	RO backs up the Main Turbine trip, reports 4KV vital buses energized, and reports SI manually initiated.		
	CRS and RO verify TRIP-1 immediate actions.		
	PO receives CRS concurrence and isolates AFW to 21 SG by closing 21AF11 and 21AF21.		
CT#1 (CT-18) Isolate feed flow into (part 1) and steam flow from			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
(part 2) 21 SG prior to a transition to SGTR-3 being required. (steam flow isolation will be performed on page 19) **Part 1** **SAT_____ UNSAT_____**			
	PO receives CRS concurrence and reduces total AFW flow to no less than 22E04 lbm/hr until at least one SG NR level is > 9%, then maintains NR levels 19-33%.		
6. 21 AFW pump trip			
	PO reports when 21 AFW pump trips, and adjusts flow from 22 and 23 AFW pumps as required based on current SG NR levels.		
Note: 21 AFW pp supplies 23/24 SGs 22 AFW pp supplies 21/22 SGs 23 AFW pump supplies all SGs			
Simulator Operator: Use **REMOTE AF20** to **OFF** 2 minutes after being dispatched to remove control power from 21 AFW pump breaker.			
	PO reports all available equipment started on SEC loading.		
	PO reports which AFW pumps are running.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	PO reports correct safeguards valve alignment.		
	RO reports 21 and 22CA330s are shut.		
	RO reports containment pressure remains less than 15 psig.		
	PO reports no indication of hi steam flow.		
	PO reports all 4KV vital buses energized.		
	RO reports ventilation systems lineups correct for current plant conditions.		
	RO reports at least 2 CCW pumps are running.		
	RO reports proper ECCS flow indicated for current plant conditions.		
	PO reports AFW status, and maintains at least 22E4 lbm/hr flow until at least one SG NR level is >9%, then maintains SG NR levels 19-33%.		
	RO reports RCP status, and RCS temperature stable at or trending to 547°.		
	RO reports both RTBs are open.		
	RO reports PZR PORVs are shut with Block valves open.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO reports PZR spray status for current plant conditions.		
	RO reports RCS pressure is >1350 psig.		
	RO maintains seal injection flow to RCPs.		
	PO reports no SGs are faulted.		
	PO reports NR or WR level is rising in 21 SG.		
	CRS transitions to EOP-SGTR-1, Steam Generator Tube Rupture.		
Note: STA will report to Control Room 10 minutes after being paged to report, and initiate monitoring of CFSTs.			
	RO maintains seal injection flow to all RCPs.		
	PO reports 21 SG is ruptured, and ensures 21MS10 is set at 1045 psig.		
	PO reports 21MS10 operating as expected for current pressure.		
	PO shuts 21MS167, and verifies 21MS7, 21MS18, and 21GB4 are shut.		
Note: These steps complete CT#1			
CT#1 (CT-18) Isolate feed flow into and steam flow from 21 SG prior to a transition to SGTR-3			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
being required. **SAT____ UNSAT____**			
	PO reports 21 SG is ruptured.		
	PO reports 23 AFW pump is not the only source of AFW.		
	PO ensures 23 AFW pump speed at minimum and trips 23 AFW pump.		
	CRS dispatches operator to shut 21MS45 if not previously performed.		
	CRS dispatches operator to shut 2SS321 sample valve.		
	PO reports 21 SG is isolated from all intact SGs.		
	PO controls AFW flow if required to maintain 21 SG NR level >9%.		
	PO reports when 21MS10 opens for pressure control @ 1045 psig.		
	PO reports 21 SG pressure is >375 psig.		
	CRS determines target temperature from Table D. (503°F for ruptured SG pressure $\geq$1000 psig.)		
	PO reports intact SGs are available for cooldown.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	PO reports steam dumps are available, and places in MS Pressure Control-Manual at 25% to perform cooldown.		
	PO bypasses Tavg when Tavg Low-Low is reached.		
	Crew monitors hottest CET temperature while continuing in procedure.		
Note: When hottest CET is <503°F, crew will stop cooldown with steps on next page.			
	RO reports power is available to both PZR PORV Stop valves, both PZR PORVs are shut, and both PZR PORV Stop valves are open.		
	RO resets SI and Phase A isolation, and reports Phase B isolation reset.		
	RO opens 21 and 22CA330.		
	PO resets each SEC and reports 230V control centers are reset.		
	RO reports RHR suction is aligned to the RWST and no RHR pump discharge flow.		
	RO stops both RHR pumps.		
	PO reports 21 SG is ruptured and 21MS167 is shut.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO shuts charging pump mini flows when RCS pressure is < 1500 psig.		
	RO reports when hottest CET is less than RCS cooldown target temp.		
	PO stops the cooldown and dumps steam to maintain RCS temp < target temp.		
CT#2 (CT-19) Establish and <u>maintain</u> an RCS temperature so that transition from SGTR-1 does not occur because temperature is too high to maintain minimum subcooling, or so low it causes transition to FRTS or FRSM. **SAT____ UNSAT____**			
	PO reports 21 SG pressure is stable or rising.		
	RO reports RCS subcooling >20 deg.		
	RO reports normal PZR spray is available.		
	RO fully opens both PZR Spray Valves.		
	RO reports PZR Spray is reducing RCS pressure.		
	Crew continues depressurization.		
7. 23 RCP trip			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Simulator Operator: Insert **RT-7** on direction from Lead Evaluator once RO reports spray effectiveness. **MALF: 23 RCP trip**			
	RO reports 23 RCP trip.		
Note: The crew is in a "do loop" when reducing RCS pressure using spray, and will return to Step 19.2			
	RO reports with loss of 23 RCP, spray is no longer effectively reducing RCS pressure.		
Simulator Operator: **IF** crew continues to use spray flow to lower RCS pressure, **THEN** insert **RT-9** on direction from Lead Evaluator to trip 21 RCP.			
	RO shuts both PZR Spray valves.		
	RO reports both PZR PORVs are available.		
	Crew reviews depressurization criteria.		
	CRS directs RO to open only one PORV.		
	IF 2PR1 is selected as the PORV to depressurize, RO reports 2PR1 will not open.		
	RO opens 2PR2.		
Simulator Operator: Insert **RT-13**			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
as soon as 2PR2 is open. **MALF: 2PR2 fails to 100%.**			
	RO reports when condition in Table E is met.		
8. PZR PORV fails to shut			
	RO reports 2PR2 will not shut.		
	CRS directs closure of 2PR7 PZR PORV Block Valve.		
Simulator Operator: Ensure **ET-3** is TRUE when 2PR7 shut PB is depressed. This fails 2PR7 90% open.			
	RO reports that after initiating closed on 2PR7, the open limit extinguished but the closed limit has not illuminated.		
	RO reports RCS pressure is NOT rising.		
	CRS transitions to EOP-SGTR-3, SGTR with LOCA – Subcooled Recovery.		
Terminate the scenario when the transition to SGTR-3 is announced.			

VI. SCENARIO REFERENCES

A. Alarm Response Procedures (Various)

B. Technical Specifications

C. Emergency Plan (ECG)

D. OP-AA-101-111-1003, Use of Procedures

E. S2.OP-IO.ZZ-0004, Power Operation

F. S2.OP-SO.DG-0003, 2C Diesel Generator Operation

G. S2.OP-ST.500-0001, Electrical Power Systems AC Sources Alignment

H. S2.OP-AB.SG-0001, Steam Generator Tube Leak

I. S2.OP-AB.LOAD-0001, Rapid Load Reduction

J. 2-EOP-TRIP-1, Reactor Trip or Safety Injection

K. 2-EOP-SGTR-1, Steam Generator Tube Rupture

ATTACHMENT 1
UNIT TWO PLANT STATUS
TODAY

MODE: 1 POWER: 89.4% RCS BORON: 1114 MWe 1080

SHUTDOWN SAFETY SYSTEM STATUS (5, 6 & DEFUELED):

NA

REACTIVITY PARAMETERS

Control Bank D is at 193 steps.
Xenon is building in at 25 pcm / hr following downpower last shift

MOST LIMITING LCO AND DATE/TIME OF EXPIRATION:

EVOLUTIONS/PROCEDURES/SURVEILLANCES IN PROGRESS:

S2.OP-PT.TRB-0003, Main Turbine Valve testing is being briefed by a dedicated test team to be performed in one hour.

ABNORMAL PLANT CONFIGURATIONS:

2C EDG Jacket Water heater is failed and C/T.

CONTROL ROOM:

Unit 1 and Hope Creek at 100% power.
No penalty minutes in the last 24 hrs.

PRIMARY:

SECONDARY:

Condensate Polisher is in service

RADWASTE:

No discharges in progress

CIRCULATING WATER/SERVICE WATER:

ATTACHMENT 2

SIMULATOR READY FOR TRAINING CHECKLIST

____ 1. Verify simulator is in "TRAIN" Load

____ 2. Simulator is in RUN

____ 3. Overhead Annunciator Horns ON

____ 4. All required computer terminals in operation

____ 5. Simulator clocks synchronized

____ 6. All tagged equipment properly secured and documented

____ 7. TSAS Status Board up-to-date

____ 8. Shift manning sheet available

____ 9. Procedures in progress open and signed-off to proper step

____ 10. All OHA lamps operating (OHA Test) and burned out lamps replaced

____ 11. Required chart recorders advanced and ON (proper paper installed)

____ 12. All printers have adequate paper AND functional ribbon

13. Required procedures clean

____ 14. Multiple color procedure pens available

____ 15. Required keys available

____ 16. Simulator cleared of unauthorized material/personnel

____ 17. All charts advanced to clean traces and chart recorders are on.

____ 18. Rod step counters correct (channel check) and reset as necessary

____ 19. Exam security set for simulator

____ 20. Ensure a current RCS Leak Rate Worksheet is placed by Aux Alarm Typewriter

____ with Baseline Data filled out

____ 21. Shift logs available if required

____ 22. Recording Media available (if applicable)

____ 23. Ensure ECG classification is correct

____ 24. Reference verification performed with required documents available

____ 25. Verify phones disconnected from plant after drill.

____ 26. Verify EGC paperwork is marked "Training Use Only" and is current revision.

____ 27. Ensure sufficient copies of ECG paperwork are available.

ATTACHMENT 3

CRITICAL TASK METHODOLOGY

In reviewing each proposed CT, the examination team assesses the task to ensure, that it is essential to safety. A task is essential to safety if, in the judgment of the examination team, the improper performance or omission of this task by a licensee will result in direct adverse consequences or in significant degradation in the mitigative capability of the plant.
The examination team determines if an automatically actuated plant system would have been required to mitigate the consequences of an individual's incorrect performance. If incorrect performance of a task by an individual necessitates the crew taking compensatory action that would complicate the event mitigation strategy, the task is safety significant.

I. Examples of CTs involving essential safety actions include those for which operation or correct performance prevents...

- degradation of any barrier to fission product release
- degraded emergency core cooling system (ECCS) or emergency power capacity
- a violation of a safety limit
- a violation of the facility license condition
- incorrect reactivity control (such as failure to initiate Emergency Boration or Standby Liquid Control, or manually insert control rods)
- a significant reduction of safety margin beyond that irreparably introduced by the scenario

II. Examples of CTs involving essential safety actions include those for which a crew demonstrates the ability to...

- effectively direct or manipulate engineered safety feature (ESF) controls that would prevent any condition described in the previous paragraph.
- recognize a failure or an incorrect automatic actuation of an ESF system or component.
- take one or more actions that would prevent a challenge to plant safety.
- prevent inappropriate actions that create a challenge to plant safety (such as an unintentional Reactor Protection System (RPS) or ESF actuation.

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

SCENARIO IDENTIFIER: 15-01 ILOT NRC ESG-2 REVIEWER: C Recchione

Initials		Qualitative Attributes
CR	1.	The scenario has clearly stated objectives in the scenario.
CR	2.	The initial conditions are realistic, in that some equipment and/or instrumentation may be out of service, but it does not cue crew into expected events.
CR	3.	The scenario consists mostly of related events.
CR	4.	Each event description consists of: • the point in the scenario when it is to be initiated • the malfunction(s) that are entered to initiate the event • the symptoms/cues that will be visible to the crew • the expected operator actions (by shift position) • the event termination point
CR	5.	No more than one non-mechanistic failure (e.g., pipe break) is incorporated into the scenario without a credible preceding incident such as a seismic event.
CR	6.	The events are valid with regard to physics and thermodynamics.
CR	7.	Sequencing/timing of events is reasonable, and allows for the examination team to obtain complete evaluation results commensurate with the scenario objectives.
CR	8.	The simulator modeling is not altered.
CR	9.	All crew competencies can be evaluated.
CR	10.	The scenario has been validated.
CR	11.	If the sampling plan indicates that the scenario was used for training during the requalification cycle, evaluate the need to modify or replace the scenario.
CR	12.	ESG-PSA Evaluation Form is completed for the scenario at the applicable facility.

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

Initial		Quantitative Attributes (as per ES-301-4, and ES-301 Section D.5.d)
GG	2	Malfunctions after EOP entry: 1-2
GG	3	Abnormal Events: 2-4
GG	1	Major Transients: 1-2
GG	1	EOPs entered/requiring substantive actions: 1-2
GG	0	EOP contingencies requiring substantive actions: 0-2
GG	2	EOP based Critical tasks: 2-3

COMMENTS:

ATTACHMENT 5

ESG CRITICAL TASKS

15-01 ILOT NRC ESG-2

CT#1 (CT-18) Isolate feed flow into and steam flow from 21 SG prior to a transition to SGTR-3 being required.

Basis: Failure to isolate the ruptured SG causes a loss of differential pressure between the ruptured SG and the intact SGs. The fact that the crew allows the differential pressure to dissipate and, as a result, are then forced to transition to a contingency ERG constitutes an incorrect performance that "necessitates the crew taking compensating action that would complicate the event mitigation strategy...."

CT #2: (CT-19) Establish and maintain an RCS temperature so that transition from SGTR does not occur because temperature is too high to maintain minimum subcooling, or so low it causes transition to FRTS or FRSM.

Basis: Failure to establish and maintain the correct RCS temperature during a SGTR leads to a transition from E-3 to a contingency ERG. This failure constitutes an incorrect performance that "necessitates the crew taking compensating action that would complicate the event mitigation strategy...."

Note to Evaluator: CT numbers in parentheses are the corresponding Westinghouse ERG Rev. 2- based Critical Tasks procedure WCAP-17711-NP

ATTACHMENT 6

ESG-PRA RELATIONSHIP EVALUATION

EVENTS LEADING TO CORE DAMAGE

Y/N	Event	Y/N	Event
N	TRANSIENTS with PCS Unavailable	N	Loss of Service Water
Y	Steam Generator Tube Rupture	N	Loss of CCW
N	Loss of Offsite Power	N	Loss of Control Air
N	Loss of Switchgear and Pen Area Ventilation	N	Station Black Out
N	LOCA		

COMPONENT/TRAIN/SYSTEM UNAVAILABILITY THAT INCREASES CORE DAMAGE FREQUENCY

Y/N	COMPONENT, SYSTEM, OR TRAIN	Y/N	COMPONENT, SYSTEM, OR TRAIN
N	Containment Sump Strainers	N	Gas Turbine
N	SSWS Valves to Turbine Generator Area	Y	Any Diesel Generator
N	RHR Suction Line valves from Hot Leg	Y	Auxiliary Feed Pump
N	CVCS Letdown line Control and Isolation Valves	N	SBO Air Compressor

OPERATOR ACTIONS IMPORTANT IN PREVENTING CORE DAMAGE

Y/N	OPERATOR ACTION
N	Restore AC power during SBO
N	Connect to gas turbine
N	Trip Reactor and RCPs after loss of component cooling system
N	Re-align RHR system for re-circulation
N	Un-isolate the available CCW Heat Exchanger
N	Isolate the CVCS letdown path and transfer charging suction to RWST
Y	Cooldown the RCS and depressurize the system
Y	Isolate the affected Steam Generator that has the tube rupture(s)
N	Early depressurize the RCS
N	Initiate feed and bleed

Complete this evaluation form for each ESG.